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We propose to study liquid helium theoretically with one of the main goals to provide support and predictions for several scheduled NASA space experiments. Advances in computer technology have enabled us to build an inexpensive massively parallel cluster with very high performance to cost ratio. In addition, we will use state of the art simulation and finite-size scaling techniques. These tools and techniques allow us to study both static as well as dynamic critical properties. We propose to study scaling of thermal conductivity near the lambda point of helium confined in planar and pore like geometries similar to those to be used in the experiment BEST. In addition, we will study the behavior of thermal conductivity near the lambda point in the presence of a heat current and we hope to be able to provide a framework to understand the results of the experiment DYNAMX. We will study microscopically the phase diagram of helium deposited on various substrates and the role of the surface van der Waals forces in scaling of confined helium. Finally we will carry out an accurate calculation of the specific heat scaling function for helium confined in a pore-like

Task Description:

geometry. This can be relevant to CHEX-II if selected for a re-flight experiment. We will use our simulations to make testable predictions for BEST, DYNAMX and MISTE that are as accurate as our predictions were for CHEX. These experiments as well as our theoretical studies of these ideal systems provide more general information about other phenomena occurring in a wide region ranging from nano-scale to micro-scale with applications to new technologies. In addition, the outcome of the proposed theoretical studies as well as the tools which will be developed as a result of this research, might have more general applications and impact in many other areas. During the past decade NASA has taken a number of key steps to support very important fundamental physics whose experimental implementation was seriously hindered by the influence of gravity. Two major experiments were carried out in space environment very successfully: the so-called Lambda Point Experiment (LPE) and the Confined Helium Experiment (CHEX). Our group funded primarily by NASA had performed a number of simulations of equilibrium critical properties such us specific heat and superfluid density of helium restricted in various geometries including the film geometry used in CHEX. Our results, published a few years before the execution of CHEX, were found to be in satisfactory agreement to the results of CHEX. We believe that that our simulations can be used for predicting the outcome of more future major experiments. In addition to testing the theoretical framework on which the simulations are based, our simulations may be used to help design the experiment.
An experiment which has been selected is the so-called BEST (Boundary Effects in the Superfluid Transition). The thermal conductivity of helium confined in pore geometry and in parallel plate geometry will be measured. This experiment tests dynamical properties with respect to scaling and the critical exponents associated with thermal conductivity. A third experiment which has been approved is the so-called DYNAMX (Critical Dynamics in Microgravity Experiment). This experiment will examine the influence of heat current on the superfluid transition temperature. Encouraged by the success of simulations of equilibrium properties we have developed techniques and computational tools to study critical dynamics and finite-size scaling of transport coefficients such as thermal conductivity using computer simulations. We propose to use these techniques to compute the thermal conductivity of superfluids in various confining geometries and investigate their finite-size scaling. In addition we shall study the shift in the Lambda transition temperature of helium in the presence of a finite-heat current (DYNAMX). In this proposal we present preliminary results for such dynamical critical phenomena relevant to BEST and DYNAMX and demonstrate that using the tools and techniques which we have developed we can perform computations which will yield support and accurate predictions for these two experiments.

We also argue that it is important to carry out the CHEX-II experiment which has been proposed by Lipa et al. This experiment is an extension of the original CHEX experiment and will measure the heat capacity of helium confined in a pore-like geometry near the lambda point. We have published preliminary results on the problem and we have pointed out that they were very different in the critical region from the results obtained for original CHEX's parallel-plate geometry. We propose to study more extensively this case using our recently developed techniques and tools (briefly outlined here) which will allow us to go far beyond the size limitations we had earlier.

Finally, we have developed a path integral Monte Carlo method to study helium (and other quantum films) on various substrates. Our previous results on this problem show a wealth of interesting phenomena and phase transitions as a function of coverage, layer and temperature. We propose to perform calculations on helium films using graphite substrate or graphite prep-lated by hydrogen. We will also study helium-3 near the gas-liquid critical point as it grows on a metallic substrate such as copper. In addition to the insight we might gain by exploring the new phenomena occurring in these films, we believe that these studies will help us understand the role of van der Waals forces on the finite-size scaling near the critical point and this is expected to be useful for MISTE. The theory of second order phase transitions is based on the assumption that at temperatures close to the critical temperature there is only one dominating length scale associated with the critical behavior of the system. This length scale is the correlation length associated with the decay of the correlation function of the order parameter. Since the correlation length diverges as the critical temperature is approached, the microscopic details of the system are irrelevant for its critical behavior. This intuitive picture has its foundation in the renormalization group treatment of second order phase transitions. Within the renormalization group treatment, it becomes evident that the critical behavior can be divided into different universality classes. Each universality class is specified by only a few key features, such as the mathematical character of the order parameter and the dimensionality of the system. Vastly different systems, for example, a superfluid and a magnetic system, can belong to the same universality class when the order parameter in both systems is a two-component vector. This implies that the set of critical exponents which describe the singular behavior of physical quantities in terms of the reduced temperature are the same. Because of this universal behavior of phase transitions, one can select an ideal and easier to work with system to study the critical behavior of all systems from each class at once. Thus, the results of this research are directly applicable to the static and transport critical properties of vastly different systems.

Phase transitions occur in nature when a macroscopic system self-organizes itself entirely to a completely different form of internal order to ``adopt' to different values of externally controlled parameters. This general phenomenon affects and occurs in a variety of different systems including chemical, physical and biological systems. The possibility for the validity of a unified theory of such phase transitions and their classification with only a few universality classes is of great importance. Furthermore, the accurate description of not only the static (equilibrium) but also the dynamic phenomena occurring in the vicinity of these transitions can have direct impact in a number of applications where external control is required. Simple examples of such processes include chemical reactions, electrical and thermal conduction, new material growth processes, etc.

Our proposal is to study confined systems, the finite-size scaling of their equilibrium and transport properties when the correlation length becomes of the size of the system or larger, the effects of their boundaries, surface effects and how these systems grow on a substrate. We will study helium as an example and because these effects, including the quantum mechanical effects, are easier to understand and demonstrate their role in a more controlled system such as helium. Many of our conclusions find applications in many other remarkably different systems because of universal behaviors and the general nature of the models used for their description. Examples of such other different systems with common or similar framework in which their description is based are other confined systems such as magnetic films, magnetic nanostructures, electrical transport in restricted geometries (with applications in nano-fabrication and nano-technology). To be more specific we give a particular example; the finite-size scaling (FSS) behavior of the low temperature electrical resistivity in thin wires (which can be currently nanofabricated with widths ranging from a few tens of nanometers to microns), in certain limits, is expected to be the same to that of the FSS behavior of thermal resistivity of helium confined in a pore geometry (the BEST experiment). Because in practice one is interested in systems of finite-size, just a bulk theory of phase transitions is not satisfactory. It is, therefore, important to study and test the FSS theory near a phase transition.

	The lessons to be learned from the theoretical (and to some extent the experimental) studies of such systems and models have applications in other areas of physics. For example, quantum field theories can be regularized by putting them on a lattice (e.g., the Lattice Gauge Theories) which represents the discretized 3+1 dimensional space-time. This allows numerical computations of properties such as the masses of sub-nuclear particles, their spectra, cross sections, etc. In order to extract values independent of the lattice cutoff a, one needs to take the so-called continuum limit of these theories. This means that the theory should have a critical behavior, just like a second order phase transition, so that a correlation length (which is a measure of the inverse of a physical mass scale) should diverge in units of the cutoff scale. This is necessary so that the correlation length to remain finite in physical units as we take the continuum limit. Thus, the entire computational material is very similar to what we use in this proposed work. Phase transitions occur not just in condensed matter physics but in other areas such as astrophysical phenomena, e.g. neutron stars. It is believed that as the universe cooled down there were a number of such phase transitions which took place at various stages of its evolution. There are speculations about a true thermodynamic phase transition which takes place in Quantum Chromodynamics, namely the transition between a quark confining phase and a phase in which the quarks and gluons are deconfined. This transition is expected to be observed in ultra-relativistic heavy ion collisions and can be described by the same theory and tools studied in this proposal.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In this funding period we used path-integral Monte Carlo (PIMC), a quantum simulation technique, to study the recently observed superfluidity in solid helium. The phase diagram of helium has been studied since the beginning of the twentieth century and was thought to be well understood. However, recently Kim and Chan reported observation of superflow in solid helium using the tortional-oscillator technique. Helium has played a key role in the development of the entire field of low temperature physics. In addition, due to the fact that the atom-atom interaction is known, it has been the playground for the development of theoretical methods to study the quantum many-body problem. We have studied the possibility that the ground state of solid Helium-4 is an incommensurate solid with a small deviation from commensurability which is very difficult to detect experimentally. If the ground state of solid helium is such a quantum incommensurate solid, it can support a superfluid state in a similar way as a striped ground state can be a superconductor in the copper-oxide materials. We used the usual PIMC method where particle permutations are sampled from a list and in addition the recently developed worm algorithm. We have studied the role of atomic vacancies introduced in an otherwise perfect HCP solid. Our inverstigations show a very interesting behavior which may be related to the supersolid phase and the detailed results will be submitted for publication soon. In addition during this funding period, we studied the effects of adding a long-range repulsive Coulomb interaction to the usual Van der Waals interaction between atoms of a submonolayer quantum film such as helium on graphite or a pure two-dimensional superfluid. Such interactions frustrate or compete with the natural tendency of the system for phase separation namely to form a macroscopic liquid or solid phase. We find that as a function of the relative strength of the long-range repulsion, surface coverage and temperature, the system undergoes a series of transformations, including a triangular Wigner-like crystal of clusters, a charge stripe-ordered phase and a fluid phase. In addition, we have investigated the existense of microemulsion phases in these systems. Such phases have been recently proposed to exist in the two-dimensional electron gas at low temperature. The goal of these studies is to understand the role of quantum fluctuations when such competing interactions appear and our results are relevant to studies of Bose-Einstein condensation and superfluidity which are funded by NASA under the laser cooling and atomic physics (LCAP) and low temperature condensed matter physics (LTCMP) programs.
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